

Memory Matters

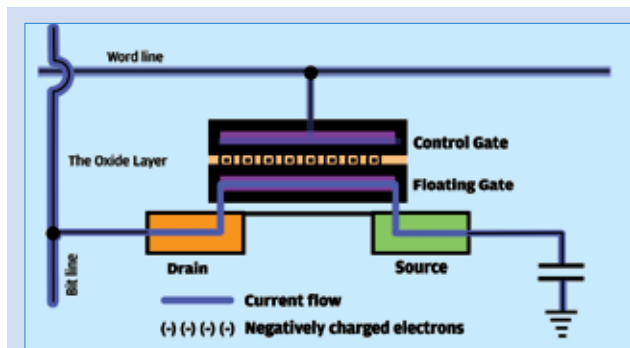
So how does Flash Memory really work?

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Most of you know that the most compact memory type is called Flash memory. Currently, Flash memory is most popularly available as USB drives, cell phone or camera memory, portable media players (PMPs), handheld consoles, and even laptops—basically anything that's built to be portable. There's even talk of Flash replacing the bulky and power-hungry hard drives that have been around for ages. Flash too, is not very young, and was invented by Dr Fujio Masuoka in 1964, while he was working for Toshiba.

Gates And Stacks

At the most basic, microscopic level, flash memory is composed of an array of floating gate transistors called cells. These cells are of two types: NAND (Not AND) and NOR (Not OR). The cells are also classified in two ways depending on how they store data—binary 0s and 1s: SLC (single level cells), which are older, can store only one bit at a time, while MLC (multi level cells) store more than one bit per cell by varying the electrical charge. Obviously, MLC devices will be able to store more data per unit area than the



A flash cell is made up of two transistors separated by a thin oxide layer. These transistors, either NAND or NOR, are called floating gates and control gates. A floating gate is connected to the array by a link called the wordline, which passes through the control gate, as shown in the figure. When the link is connected, the value of the cell is 1, which is the default state. To change the value to 0, a process called tunnelling is used. The process is simple—apply enough charge to reverse the direction of flow of electrons. Data is erased when a voltage of about 12 to 15V is applied to the blocks. When the charge of a block is over 50 per cent of this value the bit state is 1; less than 50 per cent means the bit state is 0.

legacy SLC devices. Such devices are already in the market, and though they cost more than SLC ones, research is on to cut costs and make them more available.

Unfortunately, the laws of physics set a limit on the amount of miniaturisation possible—too much, and leakage of electrons causes data loss. Multi-level cells are created by stacking two “dies” of the same size, hence doubling the capacity. This is the type of memory used in iPods. However, though MLCs double the capacity, they also reduce access speeds.

NAND vs NOR

NAND and NOR flash drives have considerable differences. Those with NOR gates provide high speed and random access to data. This means that data can be read and written to any location in the memory without having to access the memory

blocks sequentially. The minimum information that can be retrieved from a NOR flash is 1 byte. This makes NOR suitable for use in BIOS chips and for storing mobile phone firmware, where high-speeds and minute data transfers take place.

NAND, on the other hand stores and retrieves data sequentially, one block at a time. It also has a fixed lower limit below which it cannot transfer data. This data is usually what resides in an individual block, also known as pages. NAND flash memory is used for data storage and can be found in external hard drives, pen drives, memory cards and the like. NAND drives typically have higher capacities, and are cheaper.

Formats

For use as a data storage medium, NAND flash comes in several formats. The oldest is SmartMedia, with other newer standards like MultiMediaCard and CompactFlash. MultiMediaCard (MMC) has a new avatar in the form of the Secure Digital format, which itself has several component formats such as SDHC (Secure Digital High Capacity). MicroSDHC, microSD and miniSDHC. Most of these cards are used in consumer electronic devices such as cameras and mobile phones. Besides, there are several proprietary standards like Memory Stick (Sony) and xD Picture Card (Olympus, Fujifilm) that are used in products made by specific manufacturers. A newer standard, UFS (Universal Flash Storage) is being developed by a consortium for release in 2009, and will be backward-compatible with present formats.

New Hope

Manufacturers are aggressively working to increase data densities and reduce costs. A 3D flash technology is being developed that inserts silicon columns between layers, instead of trying to stack multiple layers. Although not immediately available on the shelves, you can hope to see such flash cards and devices in a year or two. ■

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